

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

4C6kBhezHE2GhCECbZRww9YcH2PNTZ3oEa1pN9Azpump

Date: 5/1/2025

Compounds: Ketamine, Oxycodone, PCP

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Ketoxycyclidine

Formula: C₂₅H₃₂ClNO₅

Weight: 460.1

Drug-likeness: 0.6

Activity:

Central Nervous System depressant and analgesic

Target Analysis

N-Methyl-D-Aspartate receptor (NMDA)

Binding Affinity: -9.5

Key role in excitatory neurotransmission; inhibition can lead to decreased pain perception and sedation.

Delta-opioid receptor

Binding Affinity: -8.7

Involved in analgesic effects and mood regulation.

Dopamine transporter

Binding Affinity: -8.9

Regulation of dopamine levels in the synaptic cleft, affecting mood and alertness.

Toxicity Profile

Risk Level: High

Affected Systems: Brain, Liver, Lungs

Mitigation Suggestions:

- Monitor dosage carefully
- Regular liver function tests
- Co-administration with respiratory stimulants in severe cases

Pharmacokinetics

Absorption:

Oral absorption is rapid but variable due to first-pass metabolism.

Distribution:

Widely distributed in the body, especially in the central nervous system.

Metabolism:

Primarily hepatic via cytochrome P450 enzymes.

Excretion:

Renal excretion of metabolites.

Half-life: 4-6 hours

Detailed Analysis

The creation of Ketoxycyclidine has leveraged the distinct pharmacological profiles of Ketamine, Oxycodone, and PCP to create a novel compound that holds potential in addressing complex clinical states of pain and mood disorders. The compound's ability to inhibit NMDA receptors is expected to potentiate its analgesic effect while the opioid receptor modulation provides a sustained pain relief mechanism. Inhibiting dopamine reuptake may offer additional mood-stabilizing properties, rendering it a candidate for treatment-resistant depression. Despite promising pharmacodynamic attributes, significant consideration must be given to its addiction potential and respiratory side effects. Tailored clinical trials focusing on safety and pharmacodynamics will define the future viability of Ketoxycyclidine.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>